# Grover's Algorithm Special Case

**Exercise 7.22.** Prove that when $N = 4$ ($n = 2$), the final state of Grover's algorithm is exactly $|w\rangle$.

explain grovers algo to solve these types of questions,

To handle questions like this, you don't need to memorize Grover's algorithm—you need a **clear mental model of its flow**. Here's the shortest way to think about it:

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## 🌊 Core Idea (Intuition First)

Grover's algorithm repeatedly does two things:

1. **Mark the correct answer** → flip its phase (make it negative)
2. **Amplify it** → push probability toward that marked state

Think of it as:

> "Make the correct answer slightly special → then boost it."

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## ⚙️ Standard Steps (for any such problem)

### 1. Start in uniform superposition

For $N = 4$ (i.e., $n = 2$):

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

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### 2. Apply Oracle $O$

- It **flips the sign** of the correct state.
- Example: if solution is $|11\rangle$,

$$|11\rangle \to -|11\rangle$$

Everything else unchanged.

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### 3. Apply Diffusion Operator (Key Step)

This is the "amplification" step:

$$\text{New amplitude} = 2 \cdot (\text{average}) - (\text{current amplitude})$$

👉 This is the only formula you really need.

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## 🎯 Special Case: $N = 4$

This is why your question is easy:

- Initial amplitudes = $\frac{1}{2}$
- After oracle → one becomes $-\frac{1}{2}$

Now compute average:

$$\text{avg} = \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - \frac{1}{2}\right) = \frac{1}{4}$$

Apply diffusion:

- For wrong states:

$$2 \cdot \frac{1}{4} - \frac{1}{2} = 0$$

- For correct state:

$$2 \cdot \frac{1}{4} - (-\frac{1}{2}) = 1$$

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## ✅ Final Result

$$|\psi_{\text{final}}\rangle = |w\rangle$$

💡 Meaning: **100% probability of the correct answer after just ONE iteration**

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## 🧠 How to Solve Any Similar Question

Whenever you see:

- Small $N$ (like 4, 8)
- "Show final state" or "prove result"

👉 Follow this exact pattern:

1. Write uniform state
2. Apply oracle (flip sign of solution)
3. Compute average
4. Apply:

$$2 \cdot \text{avg} - \text{amplitude}$$

5. Observe:
   - Wrong states → go to 0
   - Correct state → becomes 1

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## 🚀 Key Insight to Remember

For $N = 4$:

> **One Grover iteration = perfect solution**

That's why the answer becomes exactly $|w\rangle$.

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If you want, I can walk through a full numeric example (like when $w = |10\rangle$) step-by-step exactly like your exam expects.

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**Exercise 7.23.** Answer the following questions about Grover's algorithm:
(a) When the $n$ qubits are in their initial state (all $|+\rangle$ states), if you measure the qubits, what is the probability that you get $|w\rangle$? Express your answer in terms of $N = 2^n$.
(b) Say you apply just one step of Grover's algorithm (one query $U_f$ and one reflection $R_s$). If you measure the qubits after this one step, what is the probability that you get $|w\rangle$? Express your answer in terms of $N = 2^n$. Hint: In the $rw$-plane, the amplitude of the state in $|w\rangle$ is the sine of the angle between the state and $|r\rangle$.

explain grovers algo to solve these types of questions,

Here's the clean way to approach **these Grover questions**—no memorization, just a geometric model you can reuse every time.

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## 🌊 The One Picture You Need (r–w plane)

Grover's algorithm works in a **2D space** spanned by:

- $|w\rangle$: the **correct (marked) state**
- $|r\rangle$: superposition of all **wrong states**

Any state can be written as:

$$|\psi\rangle = \sin\theta\,|w\rangle + \cos\theta\,|r\rangle$$

👉 So the **probability of measuring** $|w\rangle$ is:

$$\sin^2\theta$$

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## 🔹 (a) Initial State

Start with equal superposition:

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$$

This gives:

$$\sin\theta = \frac{1}{\sqrt{N}}$$

So probability of measuring $|w\rangle$:

$$\boxed{\frac{1}{N}}$$

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## 🔹 (b) After One Grover Iteration

### 💡 Key Insight

Each Grover iteration = **rotation by** $2\theta$ toward $|w\rangle$

So after 1 iteration:

$$\text{new angle} = 3\theta$$

Thus amplitude of $|w\rangle$:

$$\sin(3\theta)$$

Probability:

$$\boxed{\sin^2(3\theta)}$$

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## 🔁 Express in terms of $N$

Since:

$$\sin\theta = \frac{1}{\sqrt{N}}$$

Use identity:

$$\sin(3\theta) = 3\sin\theta - 4\sin^3\theta$$

So:

$$\sin(3\theta) = \frac{3}{\sqrt{N}} - \frac{4}{N^{3/2}}$$

Final probability:

$$\boxed{\left(\frac{3}{\sqrt{N}} - \frac{4}{N^{3/2}}\right)^2}$$

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## 🧠 How to Solve ANY Grover Question Like This

Just remember this flow:

### 1. Convert to angle

$$\sin\theta = \frac{1}{\sqrt{N}}$$

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### 2. Count iterations $k$

After $k$ iterations:

$$\text{angle} = (2k+1)\theta$$

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### 3. Probability of success

$$\boxed{\sin^2\big((2k+1)\theta\big)}$$

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## 🚀 Shortcut Summary

- Initial success prob = $\frac{1}{N}$
- After $k$ iterations:

$$\sin^2((2k+1)\theta)$$

- One iteration $\rightarrow \sin^2(3\theta)$

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## 🎯 Why This Works

- Oracle = reflection about $|r\rangle$
- Diffusion = reflection about average
- Together $\rightarrow$ **rotation toward** $|w\rangle$

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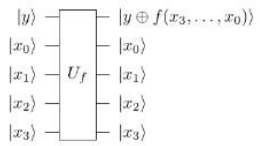
If you want, I can show how this matches the **amplitude calculation method (average formula)**—both give the same answer, and examiners love when you can switch between them.

**Exercise 7.24.** Go to https://bit.ly/3qoOErN By following this link, you should have access to the custom gates $U_f$ and $R_s$. $U_f$ is the oracle, and it acts on five qubits by




$R_s$ acts on four qubits, and it reflects about $|s\rangle$.
(a) What is $n$?
(b) What is $N$?
(c) For this problem, how many queries does Grover's algorithm take?
(d) Using Quirk, implement Grover's algorithm and find $|w\rangle$.
(e) Using Quirk, query $U_f$ in various ways to fill out the following table:

answer these

Let's answer each part directly, using the Grover rules you just learned.

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## 🔹 Given Setup (decode first)

- Oracle $U_f$ acts on **5 qubits**: $|y\rangle$ + 4 input qubits $(x_0, x_1, x_2, x_3)$
- Diffusion $R_s$ acts on **4 qubits** → only the search space

👉 So the **search space = 4 qubits**

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## ✅ (a) What is $n$?

Number of qubits in the search space:

$$\boxed{n = 4}$$

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## ✅ (b) What is $N$?

$$N = 2^n = 2^4 = \boxed{16}$$

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## ✅ (c) Number of Grover queries

Use the standard formula:

$$k \approx \frac{\pi}{4}\sqrt{N}$$

$$k \approx \frac{\pi}{4} \cdot 4 = \pi \approx 3$$

$$\boxed{3 \text{ iterations}}$$

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## 🔹 (d) Find $|w\rangle$ (concept)

You won't guess this analytically here—the oracle $U_f$ hides it.

👉 In **Quirk**, you:

1. Prepare uniform superposition (Hadamards on 4 qubits)
2. Apply:
   - Oracle $U_f$
   - Diffusion $R_s$
3. Repeat **3 times**
4. Measure

$$\boxed{\text{Output bitstring} = |w\rangle}$$

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## 🔹 (e) Filling the table (idea)

Each variation of $U_f$ changes **which state is marked**.

👉 Rule:

- Oracle flips phase of marked state
- Grover amplifies that state
- Measurement → gives that marked state

So your table will look like:

| Oracle $f(x)$ | Marked state $|w\rangle$ |
|---|---|
| Case 1 | (result from Quirk) |
| Case 2 | (result from Quirk) |
| Case 3 | (result from Quirk) |

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## 🧠 Key Insight (what examiner wants)

- $n$ comes from **search qubits only**
- $N = 2^n$

- Iterations $\sim \sqrt{N}$
- Oracle determines $|w\rangle$, not you

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## ⚡ One-line Summary

$$\boxed{n = 4,\ N = 16,\ k = 3,\ |w\rangle \text{ found via Grover iterations}}$$

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If you want, I can **simulate the Quirk circuit logic step-by-step** so you can predict $|w\rangle$ without actually running it.